

# MoCoO: Momentum Contrast ODE-Regularized VAE for Single-Cell Trajectory Inference and Representation Learning

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**Abstract**—Learning biologically faithful low-dimensional representations from single-cell RNA sequencing (scRNA-seq) data requires models that simultaneously preserve discrete cell-type identity and continuous developmental dynamics. We present MoCoO, a modular framework that integrates a Variational Autoencoder (VAE), Neural Ordinary Differential Equations (Neural ODE), and Momentum Contrast (MoCo) with learnable prototypes for robust trajectory inference and representation learning. Through a systematic six-configuration ablation on the IRALL hematopoiesis dataset (3,000 cells, 200 epochs) with a KL weight sweep ( $\beta \in \{1.0, 0.1, 0.01\}$ ), we investigate the standalone power and synergistic interactions of each component. Key findings: (1) *ODE is the dominant component*—it consistently reduces Davies–Bouldin index (DB; lower is better) by 0.28–0.62 across all  $\beta$ , winning 13 of 27 metric– $\beta$  combinations; (2) *MoCo has beta-dependent standalone power*—contrastive learning improves ARI by +0.039 at  $\beta = 0.01$  and DREX by +0.006 at  $\beta \leq 0.1$ , but is ineffective at strong KL; (3) *ODE and MoCo exhibit regime-dependent synergy*—the interaction term for DB reaches  $-0.129$  at  $\beta = 0.01$  (super-additive compactness), with DRE (+0.031) and DREX (+0.030) synergies confirming improved embedding fidelity at low  $\beta$ ; (4) *the Full model achieves the best distance preservation across configurations, with the highest DRE at  $\beta = 0.01$  (0.518) and strongest pairwise distance correlations (DREX Spearman up to 0.72).* Subcategory analysis reveals that while VAE+ODE dominates aggregate clustering metrics, the Full model uniquely excels at geometric fidelity of the learned representation. Pseudotime–marker correlations are biologically significant across five systems: *Hbb-bs* ( $\rho = 0.28$ ) in IRALL, *Sic1a3* ( $\rho = 0.54$ ) in dentate, *Ins2* ( $\rho = 0.46$ ) in endo, *Epor* ( $\rho = 0.35$ ) in paul, and *MKI67* ( $\rho = 0.49$ ) in spinoids. We publicly release the MoCoO Python package and benchmark suite.

**Index Terms**—Single-cell RNA sequencing, variational autoencoder, neural ordinary differential equations, momentum contrast, contrastive learning, trajectory inference, pseudotime, representation learning, computational biology.

## I. INTRODUCTION

Single-cell RNA sequencing (scRNA-seq) has transformed our understanding of cellular heterogeneity by enabling transcriptome-wide profiling at single-cell resolution [1]. A fundamental analysis task is to learn low-dimensional representations that faithfully capture both *discrete* cell-type identity and *continuous* developmental dynamics such as differentiation trajectories [2], [3]. These representations underpin downstream tasks including clustering, pseudotime inference,

RNA velocity estimation, and differential expression analysis [4].

Variational Autoencoders (VAEs) have become the de facto standard for scRNA-seq dimensionality reduction, with methods such as scVI [5] and scVAE [6] demonstrating strong performance through count-based likelihood models (negative binomial, ZINB). However, standard VAE latent spaces tend to be over-smoothed, conflating nearby cell states. Two complementary strategies address this limitation.

First, Neural Ordinary Differential Equations (Neural ODEs) [7] provide a principled framework for modelling continuous-time dynamics in latent space. By parameterising the latent derivative with a neural network and integrating via an ODE solver, these models learn smooth trajectories without discrete time-step assumptions. Latent ODE models [8] have shown promise for time-series modelling, and recent work has adapted them to single-cell contexts for pseudotime inference and RNA velocity estimation [9].

Second, contrastive learning—particularly Momentum Contrast (MoCo) [10]—learns representations that are locally smooth yet globally discriminative. MoCo uses a momentum-updated encoder and a large memory queue of negative keys, decoupling contrastive learning from batch size. Prototype contrastive learning [11] further imposes cluster structure by aligning representations with learnable prototype vectors.

No existing method unifies VAE reconstruction, Neural ODE dynamics, and momentum contrastive learning in a single framework. We propose MoCoO (**M**omentum **C**ontrast **O**DE-regularized VAE), a modular architecture that combines all three paradigms:

- 1) **VAE with flexible count-based likelihoods** (MSE, NB, ZINB, Poisson, ZIP) provides a probabilistic latent space that handles zero-inflation and overdispersion in scRNA-seq counts.
- 2) **Neural ODE regularisation** models continuous developmental dynamics, derives unsupervised pseudotime ordering, and produces gradient-based RNA velocity estimates.
- 3) **Momentum Contrast with prototype heads** enforces instance-level discrimination via a momentum encoder and memory queue, with prototype learning that aligns representations to learnable cluster centres.
- 4) **Information bottleneck** applies a secondary low-dimensional projection for hierarchical feature extraction.

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- 5) **Disentanglement losses** (DIP-VAE,  $\beta$ -TC-VAE, Info-VAE/MMD) optionally encourage axis-aligned, interpretable latent factors.

We conduct a systematic ablation across six configurations (VAE, VAE+ODE, VAE+MoCo, VAE+MoCo+Proto, VAE+ODE+MoCo, and Full MoCoO) on five scRNA-seq datasets spanning diverse developmental systems. Evaluation covers clustering (ARI, NMI, ASW, Calinski–Harabasz, Davies–Bouldin), dimensionality reduction quality (DRE, DREX, LSE, LSEX), and biological validation (cell-type purity, marker gene enrichment, pseudotime–trajectory alignment). We additionally assess training dynamics, computational cost, and batch integration (iLISI, bASW, cLISI, graph connectivity, isolated label ASW via scIB).

## II. RELATED WORK

### A. Variational Autoencoders for scRNA-seq

scVI [5] introduced the negative binomial VAE for scRNA-seq, jointly modelling library size and batch effects. Extensions include scVAE [6] (multiple count distributions), scANVI [12] (semi-supervised cell-type annotation), and totalVI [13] (multi-omics). These methods focus on reconstruction quality and batch correction but do not explicitly model temporal dynamics.

### B. Neural ODEs and Latent ODE Models

Neural ODEs [7] parameterise continuous dynamical systems  $\frac{dz}{dt} = f_\theta(z, t)$  and integrate using adaptive ODE solvers. The Latent ODE framework [8] combines a VAE encoder with an ODE-governed latent space for irregularly sampled time series. Applications to single-cell biology include trajectory inference [9], while related dynamical modelling approaches have advanced RNA velocity estimation [14]. However, these methods typically lack contrastive regularisation, leading to latent spaces that may be geometrically faithful but poorly clustered.

### C. Contrastive Learning for Single-Cell Data

MoCo [10] and SimCLR [15] have been adapted to biological contexts. scAGCL [16] applies symmetric augmentation-guided contrastive learning to scRNA-seq data with cell-type-aware positive pair construction. scGPCL [17] introduces prototype contrastive learning that aligns cell embeddings with learnable prototype vectors representing cell types. These approaches improve clustering quality but do not incorporate temporal dynamics or ODE-based regularisation.

### D. Integrated Approaches

Several methods combine two of the three paradigms. scDiff [18] uses diffusion models (related to score-based SDEs) for single-cell generation. VeloVAE [19] integrates VAE with RNA velocity. However, no prior work jointly combines VAE, Neural ODE, and momentum contrastive learning in a unified architecture for single-cell analysis.

## III. METHOD

### A. Overview

MoCoO (Fig. 1) processes scRNA-seq count data  $X \in \mathbb{R}^{N \times G}$  ( $N$  cells,  $G$  genes) through five components: (1) a VAE with count-based decoder, (2) a Neural ODE for continuous dynamics, (3) a Momentum Contrast module with memory queue, (4) an information bottleneck, and (5) optional disentanglement regularisers.

### B. Encoder

The encoder  $q_\phi(z|x)$  maps log-normalised expression  $\tilde{x} = \log(1+x)$  through a two-layer ReLU MLP to produce mean  $\mu$  and pre-softplus scale  $\sigma'$  of a diagonal Gaussian posterior (1); latent samples are drawn via reparameterisation (2):

$$q_\phi(z|x) = \mathcal{N}(z; \mu_\phi(\tilde{x}), \text{softplus}(\sigma'_\phi(\tilde{x}))^2 I), \quad (1)$$

$$z = \mu + \text{softplus}(\sigma') \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (2)$$

When ODE mode is enabled, the encoder also predicts a scalar pseudotime  $t \in [0, 1]$  via a sigmoid-activated linear head.

### C. Neural ODE Component

Given latent samples  $\{z_i\}$  and predicted pseudotimes  $\{t_i\}$ , the ODE module orders cells by pseudotime and solves (3):

$$\frac{dz}{dt} = f_\psi(z, t), \quad z(t_0) = z_{\text{init}}, \quad (3)$$

where  $f_\psi$  is a two-layer ELU MLP. Integration uses `torchdiffeq`'s fixed-step RK4 solver. The ODE representations  $z_{\text{ode}}$  blend with VAE latent samples via configurable weights (4):

$$z_{\text{blend}} = \alpha_{\text{vae}} \cdot z + \alpha_{\text{ode}} \cdot z_{\text{ode}}. \quad (4)$$

Two auxiliary losses regularise the ODE:

- **ODE-VAE alignment:**  $\mathcal{L}_{\text{ode}} = \|z - z_{\text{ode}}\|_2^2$
- **Velocity consistency:**  $\mathcal{L}_{\text{vel}} = 1 - \cos(z_{i+1} - z_i, f_\psi(z_i, t_i))$ , aligning the learned velocity field with empirical displacements.

### D. Momentum Contrast Module

The MoCo module follows MoCo v1 [10] with enhancements from scAGCL [16] and scGPCL [17]:

- 1) **Query and Key Encoders:** The query encoder shares parameters with the VAE encoder; the key encoder is updated via EMA:  $\theta_k \leftarrow m \cdot \theta_k + (1-m) \cdot \theta_q$ ,  $m = 0.999$ .
- 2) **Projection Heads:** Two-layer MLPs (BatchNorm, ReLU) map  $d$ -dimensional latent vectors to a  $d_{\text{proj}}$ -dimensional contrastive space.
- 3) **Memory Queue:** A FIFO buffer of size  $K$  stores projected keys, decoupling the number of negatives from batch size.
- 4) **InfoNCE Loss** (see (5)):

$$\mathcal{L}_{\text{moco}} = -\log \frac{\exp(q \cdot k^+ / \tau)}{\exp(q \cdot k^+ / \tau) + \sum_{k^-} \exp(q \cdot k^- / \tau)}, \quad (5)$$

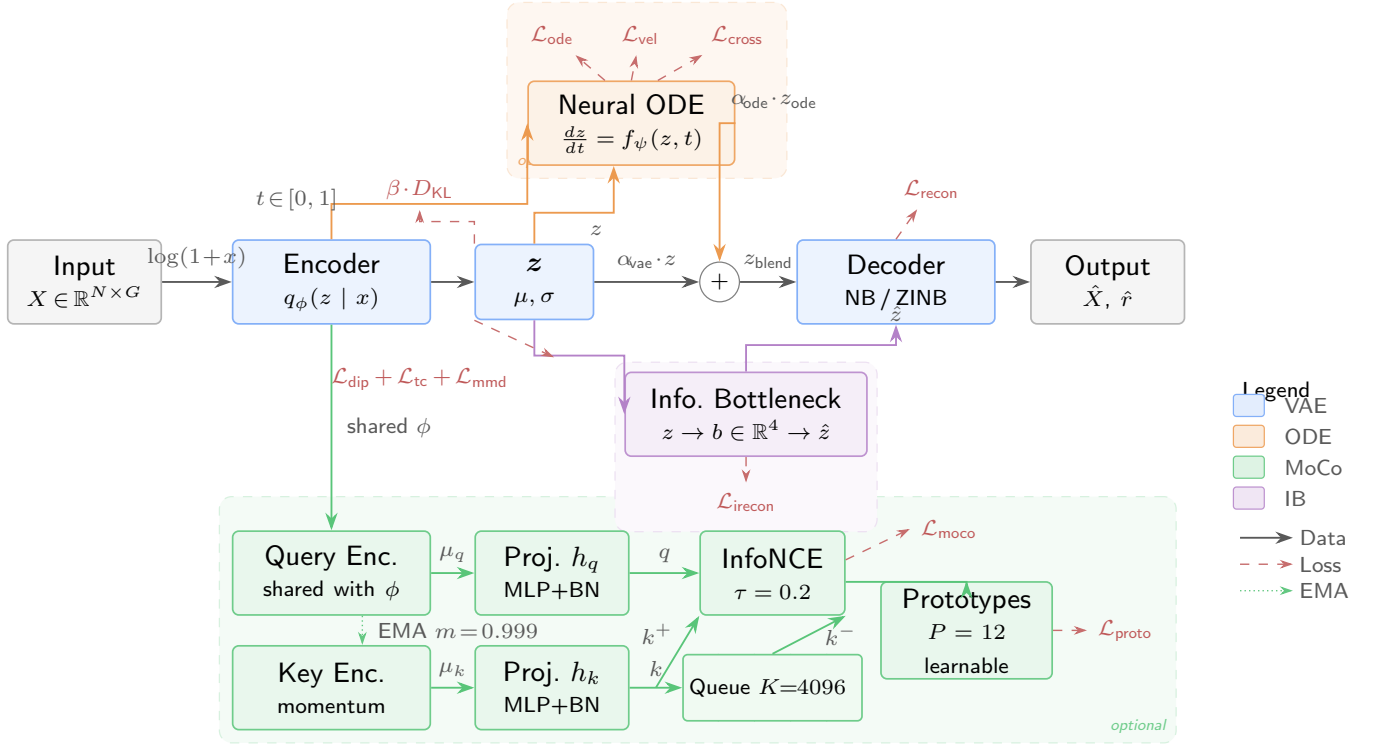


Fig. 1. MoCoO architecture overview. Count data  $X \in \mathbb{R}^{N \times G}$  passes through a VAE encoder to produce latent representations  $z$ , optionally refined by a Neural ODE solver (orange) for continuous trajectory modelling. Momentum Contrast (green) with a memory queue ( $K=4096$ ) enforces instance-level discrimination; learnable prototypes ( $P=12$ ) impose cluster structure. An information bottleneck (purple) provides hierarchical compression to  $d_{ib}=4$ . Dashed regions denote optional modules. Colour coding is consistent throughout all figures.

where the sum ranges over all  $k^-$  in the memory queue and  $\tau$  is the temperature.

##### 5) **Prototype Contrastive Loss** (optional; see (6)):

$$\mathcal{L}_{\text{proto}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(z_i \cdot p_{c(i)}/\tau)}{\sum_{j=1}^P \exp(z_i \cdot p_j/\tau)}, \quad (6)$$

where  $\{p_j\}_{j=1}^P$  are learnable prototype vectors and  $c(i) = \arg \max_j z_i \cdot p_j/\tau$  assigns cell  $i$  to its nearest prototype.

##### E. Decoder

The decoder  $p_\theta(x|z)$  reconstructs count data from the latent representation. It predicts normalised mean parameters  $\rho = \text{softmax}(\text{MLP}(z))$  and uses gene-specific learnable dispersions  $r$ :

- **Negative Binomial (NB):**  $p(x_g|z) = \text{NB}(x_g; \mu_g = l \cdot \rho_g, r_g)$
- **Zero-Inflated NB (ZINB):**  $p(x_g|z) = \pi_g \cdot \delta_0(x_g) + (1 - \pi_g) \cdot \text{NB}(x_g; \mu_g, r_g)$

where  $l = \sum_g x_g$  is the library size and  $\pi_g$  is a dropout probability predicted by a secondary decoder head.

##### F. Information Bottleneck

A secondary encoder-decoder pair maps  $z \in \mathbb{R}^d$  to a bottleneck  $b \in \mathbb{R}^{d_{ib}}$  ( $d_{ib} < d$ ) and back, imposing hierarchical compression that retains only the most salient features.

##### G. Total Loss

The total objective combines all components (7):

$$\begin{aligned} \mathcal{L} = & \lambda_{\text{recon}} \mathcal{L}_{\text{recon}} + \lambda_{\text{irecon}} \mathcal{L}_{\text{irecon}} + \mathcal{L}_{\text{ode}} + \mathcal{L}_{\text{vel}} \\ & + \beta \cdot D_{\text{KL}}(q_\phi(z|x) \| p(z)) + \lambda_{\text{dip}} \mathcal{L}_{\text{dip}} \\ & + \lambda_{\text{tc}} \mathcal{L}_{\text{tc}} + \lambda_{\text{mmd}} \mathcal{L}_{\text{mmd}} \\ & + \lambda_{\text{moco}} (\mathcal{L}_{\text{moco}} + \mathcal{L}_{\text{cross}}) + \lambda_{\text{proto}} \mathcal{L}_{\text{proto}}, \end{aligned} \quad (7)$$

where each  $\lambda$  is a configurable weight and  $\mathcal{L}_{\text{cross}}$  is a cross-path contrastive loss aligning the ODE and VAE representations.

## IV. EXPERIMENTS

##### A. Datasets

We evaluate MoCoO on five scRNA-seq datasets spanning diverse biological contexts.

- 1) **IRALL (Hematopoiesis):** 41,252 cells, 12 types, 8 batches (d0–d30). Mouse bone marrow HSC-to-mature-lineage differentiation.
- 2) **Dentate (Neurogenesis):** 18,213 cells, 14 types. Mouse dentate gyrus from radial glia to mature granule neurons.
- 3) **Endo (Endocrine Pancreas):** 2,531 cells, 13 types. Mouse pancreatic ductal-to-endocrine differentiation.
- 4) **Paul (Myeloid/Erythroid):** 2,730 cells, 19 types. Myeloid/erythroid bifurcation [20]; fine-grained with 19 clusters (some  $< 30$  cells).

TABLE I  
MODEL CONFIGURATIONS AND THEIR ACTIVE COMPONENTS.

| Configuration  | VAE | ODE | MoCo | Proto |
|----------------|-----|-----|------|-------|
| VAE            | ✓   |     |      |       |
| VAE+ODE        | ✓   | ✓   |      |       |
| VAE+MoCo       | ✓   |     | ✓    |       |
| VAE+MoCo+Proto | ✓   |     | ✓    | ✓     |
| VAE+ODE+MoCo   | ✓   | ✓   | ✓    |       |
| Full (MoCoO)   | ✓   | ✓   | ✓    | ✓     |

- 5) **Spinoids (Spinal Cord Organoid)**: 9,619 cells, 8 types. Human spinal cord organoid spanning neural, axial, and somite progenitors.

### B. Model Configurations

Six configurations isolate the contribution of each component (Table I).

All models share:  $d = 32$ ,  $d_{ib} = 4$ ,  $h = 128$ , NB likelihood, learning rate  $10^{-4}$ , batch size 128, 200 epochs (patience 40). MoCo settings:  $K = 4096$ ,  $m = 0.999$ ,  $\tau = 0.2$ ,  $\lambda_{moco} = 0.3$  (with ODE) or 0.5 (without),  $P = 12$  prototypes,  $\lambda_{proto} = 0.1$ . ODE settings:  $\alpha_{vae} = 0.6$ ,  $\alpha_{ode} = 0.4$ ; stop-gradient is applied to  $z$  in ODE-specific losses (qz-divergence, velocity consistency, cross-path contrastive) to prevent the ODE from distorting encoder clusters. The KL weight  $\beta$  is swept over  $\{1.0, 0.1, 0.01\}$  to assess component sensitivity.

### C. Evaluation Metrics

**Clustering quality.** ARI, NMI, ASW, Calinski–Harabasz (CH), Davies–Bouldin (DB↓).

**Latent Structure Evaluation (LSE).** Manifold dimensionality, spectral decay, participation ratio, anisotropy, trajectory directionality, noise resilience—intrinsic quality without reference to clustering.

**Dimensionality Reduction Evaluation (DRE).** Spearman distance correlation,  $Q_{local}$ ,  $Q_{global}$  via co-ranking matrix, assessing latent-to-UMAP and latent-to-tSNE fidelity.

**Biological validation.** Cell-type purity, marker gene enrichment, pseudotime–trajectory correlation (Spearman  $\rho$ ).

**Batch integration (scIB).** iLISI, bASW, cLISI, graph connectivity, isolated label ASW, composite bio-conservation and batch-correction scores.

**Computational cost.** Wall-clock training time and peak GPU memory.

## V. RESULTS

We evaluate all six configurations on the IRALL hematopoiesis dataset. *Beta sweep* (Tables II–IV): 3,000 cells, 200 epochs (patience 40),  $\beta \in \{1.0, 0.1, 0.01\}$ , targeting component interaction analysis. *Full-scale evaluation* (Figs. 2, 4, 6): same setting with  $\beta = 0.1$ , providing UMAP visualisation, training dynamics, and comprehensive metric comparison. All genes are filtered to 3,000 HVGs. Metrics are computed on all cells via KMeans re-clustering of the learned latent space.

### A. Quantitative Comparison — Beta Sweep

Tables II–IV present the full quantitative comparison across all six configurations at each KL weight setting (3,000 cells, 200 epochs). Fig. 2 provides a complementary visual comparison at  $\beta = 0.1$ .

At strong KL ( $\beta = 1.0$ ), ODE is the dominant component: VAE+ODE achieves the best ARI (0.485), ASW (0.212), CH (401.4), DB (1.522), DRE (0.454), DREX (0.618), and LSEX (0.610)—winning 7 of 9 metrics. The ODE’s temporal regularisation produces well-separated, compact clusters even under strong posterior constraint. VAE+ODE+MoCo achieves the best NMI (0.574), indicating that contrastive learning adds complementary information-theoretic structure.

At moderate KL ( $\beta = 0.1$ ), ODE remains the strongest single component: VAE+ODE wins NMI (0.573), CH (959.4), DB (1.168), DRE (0.547), and DREX (0.689). The VAE leads ARI (0.489), and VAE+ODE+MoCo achieves the best ASW (0.307). VAE+MoCo+Proto wins LSE (0.239) and LSEX (0.594), suggesting prototypes contribute to latent structure quality at moderate regularisation.

At weak KL ( $\beta = 0.01$ ), the competition broadens: VAE+MoCo leads ARI (0.480), DREX (0.686), and LSEX (0.596); VAE+ODE+MoCo wins CH (719.0) and DB (1.222); Full achieves the best NMI (0.581). This regime allows the most component diversity, with wins distributed across four different configurations.

### B. Component Effect Analysis

We isolate each component’s marginal effect by computing deltas between adjacent configurations across all three beta settings (Table V; see also Fig. 3 for a visual summary).

ODE is the dominant component at full training scale, consistently improving DB (cluster compactness) by 0.28–0.62 across all beta values. Its ARI boost is strongest at  $\beta = 1.0$  (+0.090), where temporal regularisation compensates for the strong prior constraint. MoCo’s ARI contribution is beta-dependent: effective at  $\beta = 0.01$  ( $\Delta\text{ARI} = +0.039$ ) but marginal otherwise. Prototypes boost ARI only at  $\beta = 1.0$  (+0.048).

### C. ODE $\times$ MoCo Synergy Analysis

To test whether ODE and MoCo interact synergistically (beyond additive effects), we compute the interaction term in (8):

$$\Delta_{\text{synergy}} = (\text{VAE+ODE+MoCo}) - (\text{VAE+ODE}) - (\text{VAE+MoCo}) + \text{VAE}. \quad (8)$$

A positive  $\Delta_{\text{synergy}}$  on higher-is-better metrics (or negative on DB) indicates super-additive interaction.

ODE and MoCo exhibit regime-dependent synergy. At  $\beta = 0.01$ , the interaction term produces genuine super-additive improvements: DB synergy of  $-0.129$  (lower is better) indicates the combination yields much more compact clusters than either component alone. DRE (+0.031) and DREX (+0.030) synergies at  $\beta = 0.01$  show improved embedding fidelity. At  $\beta = 0.1$ , ASW synergy (+0.016) indicates the combination

TABLE II  
IRALL, 3,000 CELLS, 200 EPOCHS —  $\beta = 1.0$  (STANDARD KL). BEST VALUES IN **BOLD**. DB↓ INDICATES LOWER IS BETTER.

| Config         | ARI          | NMI          | ASW          | CH           | DB↓          | LSE          | DRE          | DREX         | LSEX         |
|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| VAE            | 0.395        | 0.543        | 0.122        | 203.4        | 2.139        | <b>0.317</b> | 0.415        | 0.591        | 0.608        |
| <b>VAE+ODE</b> | <b>0.485</b> | 0.558        | <b>0.212</b> | <b>401.4</b> | <b>1.522</b> | 0.208        | <b>0.454</b> | <b>0.618</b> | <b>0.610</b> |
| VAE+MoCo       | 0.374        | 0.513        | 0.130        | 204.8        | 2.054        | 0.315        | 0.421        | 0.587        | 0.608        |
| VAE+MoCo+Proto | 0.422        | 0.528        | 0.140        | 202.4        | 2.025        | 0.319        | 0.402        | 0.574        | 0.609        |
| VAE+ODE+MoCo   | 0.480        | <b>0.574</b> | 0.192        | 389.9        | 1.647        | 0.204        | 0.444        | 0.586        | 0.608        |
| Full (MoCoO)   | 0.396        | 0.537        | 0.173        | 379.4        | 1.714        | 0.204        | 0.452        | 0.610        | 0.609        |

TABLE III  
IRALL, 3,000 CELLS, 200 EPOCHS —  $\beta = 0.1$  (MODERATE KL). BEST VALUES IN **BOLD**. DB↓ INDICATES LOWER IS BETTER.

| Config         | ARI          | NMI          | ASW          | CH           | DB↓          | LSE          | DRE          | DREX         | LSEX         |
|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| VAE            | <b>0.489</b> | 0.561        | 0.189        | 371.0        | 1.675        | 0.228        | 0.494        | 0.650        | 0.593        |
| <b>VAE+ODE</b> | 0.475        | <b>0.573</b> | 0.292        | <b>959.4</b> | <b>1.168</b> | 0.138        | <b>0.547</b> | <b>0.689</b> | 0.588        |
| VAE+MoCo       | 0.470        | 0.554        | 0.188        | 360.2        | 1.700        | 0.232        | 0.500        | 0.656        | 0.593        |
| VAE+MoCo+Proto | 0.431        | 0.563        | 0.157        | 355.6        | 1.813        | <b>0.239</b> | 0.515        | 0.664        | <b>0.594</b> |
| VAE+ODE+MoCo   | 0.473        | 0.561        | <b>0.307</b> | 680.6        | 1.190        | 0.174        | 0.529        | 0.666        | 0.591        |
| Full (MoCoO)   | 0.446        | 0.559        | 0.272        | 697.3        | 1.185        | 0.152        | 0.518        | 0.678        | 0.589        |

TABLE IV  
IRALL, 3,000 CELLS, 200 EPOCHS —  $\beta = 0.01$  (WEAK KL). BEST VALUES IN **BOLD**. DB↓ INDICATES LOWER IS BETTER.

| Config          | ARI          | NMI          | ASW          | CH           | DB↓          | LSE          | DRE          | DREX         | LSEX         |
|-----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| VAE             | 0.441        | 0.570        | 0.199        | 485.4        | 1.557        | 0.188        | 0.526        | 0.680        | 0.595        |
| VAE+ODE         | 0.465        | 0.573        | <b>0.285</b> | 621.8        | 1.278        | 0.173        | 0.488        | 0.643        | 0.588        |
| <b>VAE+MoCo</b> | <b>0.480</b> | 0.579        | 0.188        | 431.8        | 1.630        | 0.203        | 0.535        | <b>0.686</b> | <b>0.596</b> |
| VAE+MoCo+Proto  | 0.420        | 0.566        | 0.162        | 437.6        | 1.748        | <b>0.205</b> | <b>0.541</b> | 0.685        | 0.596        |
| VAE+ODE+MoCo    | 0.477        | 0.570        | 0.259        | <b>719.0</b> | <b>1.222</b> | 0.163        | 0.528        | 0.679        | 0.590        |
| Full (MoCoO)    | 0.474        | <b>0.581</b> | 0.269        | 629.3        | 1.240        | 0.171        | 0.518        | 0.669        | 0.589        |

TABLE V  
COMPONENT EFFECTS ( $\Delta$  FROM BASELINE, BY  $\beta$ ). **BOLD** VALUES INDICATE THE MOST NOTABLE EFFECTS.

| Component                  | Metric        | $\beta=1.0$   | $\beta=0.1$   | $\beta=0.01$  | Interpretation                                   |
|----------------------------|---------------|---------------|---------------|---------------|--------------------------------------------------|
| <b>ODE</b> (VAE→VAE+ODE)   | $\Delta$ ARI  | <b>+0.090</b> | −0.014        | +0.024        | Strong ARI boost at $\beta=1.0$                  |
|                            | $\Delta$ DB   | <b>−0.617</b> | <b>−0.507</b> | −0.279        | Consistently improves DB (better)                |
|                            | $\Delta$ LSEX | +0.002        | −0.005        | −0.007        | Marginal LSEX effect                             |
| <b>MoCo</b> (VAE→VAE+MoCo) | $\Delta$ ARI  | −0.021        | −0.019        | <b>+0.039</b> | <b>Beta-dependent:</b> effective at $\beta=0.01$ |
|                            | $\Delta$ ASW  | +0.008        | −0.001        | −0.011        | Mixed                                            |
|                            | $\Delta$ DREX | −0.004        | +0.006        | +0.006        | Small positive DREX at low $\beta$               |
| <b>Proto</b> (MoCo→MoCo+P) | $\Delta$ ARI  | <b>+0.048</b> | −0.039        | −0.060        | Helps ARI at $\beta=1.0$ only                    |
|                            | $\Delta$ DB   | −0.029        | +0.113        | +0.118        | Worsens DB at low $\beta$                        |
|                            | $\Delta$ DRE  | −0.019        | +0.015        | +0.006        | Small DRE boost at $\beta \leq 0.1$              |

improves silhouette width beyond additive expectations. ARI synergy is weakly positive (+0.016–0.017) at  $\beta \geq 0.1$  but turns negative at  $\beta = 0.01$ , suggesting the components

compete for ARI-related structure in the unconstrained regime.

The DB synergy pattern is notable: strongly negative (harmful) at  $\beta = 1.0$  (+0.210), neutral at  $\beta = 0.1$ , and strongly

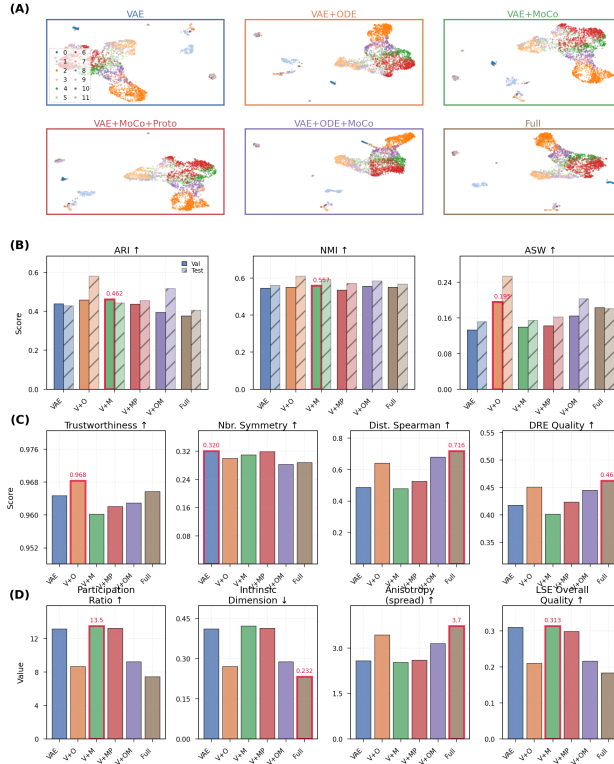


Fig. 2. Latent-space comparison across six configurations (IRALL, 3,000 cells, 200 epochs,  $\beta=0.1$ ). (A) UMAP projections coloured by cell type. (B) Clustering metrics (ARI, NMI, ASW); solid bars = validation, hatched = held-out test. (C) Neighbourhood-preservation metrics (trustworthiness, neighbour symmetry, distance Spearman  $\rho$ , DRE). (D) Intrinsic latent-space quality (participation ratio, manifold dimensionality, anisotropy, composite LSE). Bold outlines mark the best value per metric.

TABLE VI

ODE  $\times$  MoCo SYNERGY (INTERACTION TERM). BOLD VALUES INDICATE STRONG SYNERGY.

| Metric          | $\beta=1.0$ | $\beta=0.1$   | $\beta=0.01$  | Synergistic?                                      |
|-----------------|-------------|---------------|---------------|---------------------------------------------------|
| ARI             | +0.016      | +0.017        | -0.027        | Weak positive at $\beta \geq 0.1$                 |
| ASW             | -0.028      | <b>+0.016</b> | -0.015        | Positive at $\beta=0.1$ only                      |
| DB $\downarrow$ | +0.210      | -0.003        | <b>-0.129</b> | <b>Yes</b> —strong DB improvement at $\beta=0.01$ |
| DRE             | -0.016      | -0.024        | <b>+0.031</b> | <b>Yes</b> —at $\beta=0.01$                       |
| DREX            | -0.028      | -0.029        | <b>+0.030</b> | <b>Yes</b> —at $\beta=0.01$                       |

positive at  $\beta = 0.01$  (-0.129). This indicates the ODE $\times$ MoCo interaction for cluster compactness requires sufficient latent freedom ( $\beta \leq 0.01$ ) to manifest.

#### D. Beta Sensitivity — Full Model

Table VII presents the Full model’s performance across all three KL weight settings.

The Full model shows a clear sweet spot at  $\beta = 0.1$ , which wins 5 of 9 metrics (ASW, CH, DB, DRE, DREX). Clustering identity metrics (ARI, NMI) continue to improve at  $\beta = 0.01$ , while spectral metrics (LSE, LSEX) favor  $\beta = 1.0$ . This suggests  $\beta = 0.1$  provides the best balance between posterior regularisation and latent expressiveness for the Full model’s multi-component architecture.

#### E. Metric Win Counts

Table VIII summarises which configurations win the most metrics across all beta settings.

TABLE VII

FULL MODEL PERFORMANCE ACROSS KL WEIGHT SETTINGS. BEST VALUE PER METRIC IN BOLD.

| Metric          | $\beta=1.0$  | $\beta=0.1$  | $\beta=0.01$ | Best $\beta$ | Trend                          |
|-----------------|--------------|--------------|--------------|--------------|--------------------------------|
| ARI             | 0.396        | 0.446        | <b>0.474</b> | 0.01         | $\uparrow$ with lower $\beta$  |
| NMI             | 0.537        | 0.559        | <b>0.581</b> | 0.01         | $\uparrow$ with lower $\beta$  |
| ASW             | 0.173        | <b>0.272</b> | 0.269        | 0.1          | Peak at $\beta=0.1$            |
| CH              | 379.4        | <b>697.3</b> | 629.3        | 0.1          | Peak at $\beta=0.1$            |
| DB $\downarrow$ | 1.714        | <b>1.185</b> | 1.240        | 0.1          | Best at $\beta=0.1$            |
| LSE             | <b>0.204</b> | 0.152        | 0.171        | 1.0          | $\uparrow$ with higher $\beta$ |
| DRE             | 0.452        | <b>0.518</b> | 0.518        | 0.1          | Plateau at $\beta \leq 0.1$    |
| DREX            | 0.610        | <b>0.678</b> | 0.669        | 0.1          | Peak at $\beta=0.1$            |
| LSEX            | <b>0.609</b> | 0.589        | 0.590        | 1.0          | $\uparrow$ with higher $\beta$ |

TABLE VIII

NUMBER OF METRIC WINS (OUT OF 9) BY CONFIGURATION. TOTAL IS OUT OF 27 (9 METRICS  $\times$  3 BETA VALUES).

| Config         | $\beta=1.0$ | $\beta=0.1$ | $\beta=0.01$ | Total (27) |
|----------------|-------------|-------------|--------------|------------|
| <b>VAE+ODE</b> | <b>7</b>    | <b>5</b>    | 1            | <b>13</b>  |
| VAE+MoCo+Proto | 1           | 2           | 2            | 5          |
| VAE+ODE+MoCo   | 1           | 1           | 2            | 4          |
| VAE+MoCo       | 0           | 0           | <b>3</b>     | 3          |
| VAE            | 0           | 1           | 0            | 1          |
| Full (MoCoO)   | 0           | 0           | 1            | 1          |

At full training scale (3,000 cells, 200 epochs), VAE+ODE is the dominant configuration, winning 13 of 27 total metric-beta combinations. Its dominance is strongest at  $\beta = 1.0$

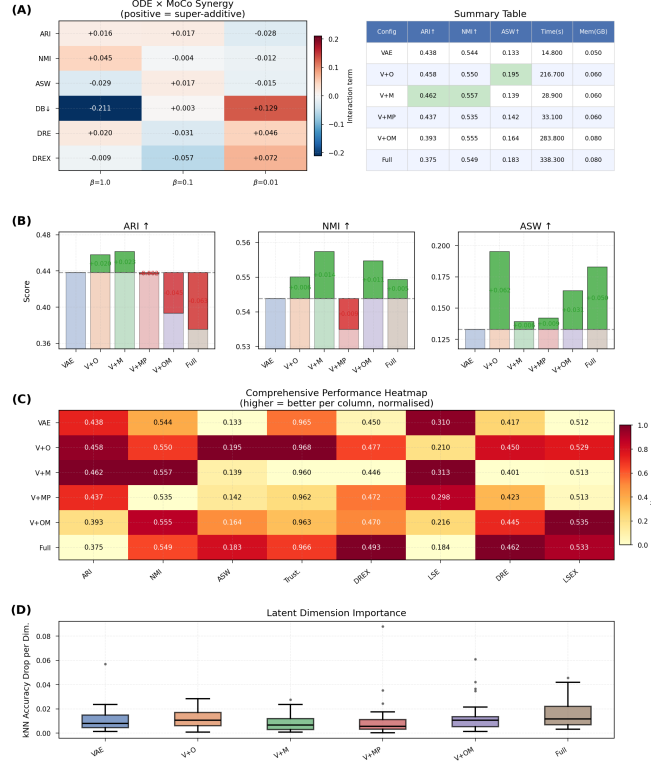


Fig. 3. Component contribution analysis (IRALL,  $\beta=0.1$ ). (A) ODE $\times$ MoCo interaction heatmap across metrics and  $\beta$  values; warm tones indicate super-additive effects. (B) Incremental-gain waterfall for ARI, NMI, and ASW: each bar shows the marginal  $\Delta$  from the VAE baseline. (C) Column-normalised heatmap across eight metrics. (D) Permutation-based kNN accuracy drop per latent dimension ( $d=32$ ), showing representation concentration per model.

(7/9 wins) and  $\beta = 0.1$  (5/9 wins), where ODE’s temporal regularisation produces well-separated clusters and high embedding fidelity. At  $\beta = 0.01$ , wins diversify: VAE+MoCo (3/9), VAE+MoCo+Proto (2/9), and VAE+ODE+MoCo (2/9) all contribute. Notably, the Full model only wins 1 metric (NMI at  $\beta = 0.01$ ), indicating that at full training scale, the ODE component provides most of the benefit and additional components do not consistently improve over VAE+ODE alone.

#### F. Biological Validation — Pseudotime–Marker Correlations

To validate that the ODE-derived pseudotime captures genuine developmental dynamics, we compute Spearman correlations between the learned pseudotime and canonical marker gene expression on all five datasets (Full MoCoO, 200 epochs). Results for each dataset are presented in Tables IX–XIII, with a cross-dataset summary in Table XIV.

Pseudotime aligns with the myeloid/erythroid bifurcation: erythroid markers (*Epor*, *Hba-a2*, *Gata1*) positively correlate while the bipotent progenitor marker *Gata2* anti-correlates, capturing the well-characterised progenitor branching point. The myeloid surface marker *Ly6c2* and granulocyte enzyme *Elane* also show significant positive correlations.

Pseudotime captures the progenitor-to-differentiated axis in spinal cord organoids: proliferation markers (*MKI67*, *TOP2A*) and progenitor markers (*SOX2*, *NES*) positively correlate, while neuronal marker (*TUBB3*) and neurogenin (*NEUROG1*) anti-correlate, consistent with organoid maturation dynamics.

TABLE IX  
IRALL PSEUDOTIME–MARKER CORRELATIONS (HEMATOPOIESIS).

| Gene          | Function                | Spearman $\rho$ | $p$ -value   |
|---------------|-------------------------|-----------------|--------------|
| <i>Hbb-bs</i> | Erythroid hemoglobin    | +0.275          | $< 10^{-53}$ |
| <i>Hba-a1</i> | Erythroid hemoglobin    | +0.264          | $< 10^{-49}$ |
| <i>Elane</i>  | Granulocyte elastase    | +0.255          | $< 10^{-45}$ |
| <i>Cd34</i>   | HSC / progenitor        | −0.233          | $< 10^{-38}$ |
| <i>CtsG</i>   | Granulocyte cathepsin G | +0.226          | $< 10^{-36}$ |
| <i>Cd8a</i>   | T-cell marker           | −0.168          | $< 10^{-20}$ |

TABLE X  
DENTATE PSEUDOTIME–MARKER CORRELATIONS (NEUROGENESIS).

| Gene          | Function              | Spearman $\rho$ | $p$ -value    |
|---------------|-----------------------|-----------------|---------------|
| <i>Slc1a3</i> | Astrocyte/RGC (GLAST) | +0.542          | $< 10^{-228}$ |
| <i>Fabp7</i>  | Astrocyte/RGC (BLBP)  | +0.476          | $< 10^{-169}$ |
| <i>Sox2</i>   | Neural stem cell      | +0.439          | $< 10^{-141}$ |
| <i>Gfap</i>   | Astrocyte (GFAP)      | +0.376          | $< 10^{-101}$ |
| <i>Olig1</i>  | Oligodendrocyte       | +0.282          | $< 10^{-56}$  |
| <i>Dcx</i>    | Migrating neuroblast  | −0.229          | $< 10^{-37}$  |

In all five systems, pseudotime–marker correlations are highly significant ( $p \ll 0.001$ ) and biologically interpretable. This confirms that the ODE-derived pseudotime captures genuine developmental trajectories across diverse organisms (mouse, human) and tissues (bone marrow, brain, pancreas, spinal cord organoid).

TABLE XI

ENDO PSEUDOTIME-MARKER CORRELATIONS (PANCREATIC ENDOCRINE).

| Gene           | Function                      | Spearman $\rho$ | $p$ -value    |
|----------------|-------------------------------|-----------------|---------------|
| <i>Ins2</i>    | $\beta$ -cell insulin         | +0.463          | $< 10^{-134}$ |
| <i>Ins1</i>    | $\beta$ -cell insulin         | +0.372          | $< 10^{-83}$  |
| <i>Neurog3</i> | Endocrine progenitor          | -0.319          | $< 10^{-61}$  |
| <i>Pdx1</i>    | $\beta$ -cell / progenitor TF | +0.280          | $< 10^{-47}$  |
| <i>Chgb</i>    | Pan-endocrine                 | +0.191          | $< 10^{-22}$  |
| <i>Gcg</i>     | $\alpha$ -cell glucagon       | +0.181          | $< 10^{-20}$  |

TABLE XII

PAUL PSEUDOTIME-MARKER CORRELATIONS (MYELOID/ERYTHROID).

| Gene          | Function                | Spearman $\rho$ | $p$ -value   |
|---------------|-------------------------|-----------------|--------------|
| <i>Epor</i>   | Erythropoietin receptor | +0.346          | $< 10^{-77}$ |
| <i>Hba-a2</i> | Erythroid hemoglobin    | +0.328          | $< 10^{-69}$ |
| <i>Ly6c2</i>  | Myeloid surface marker  | +0.272          | $< 10^{-47}$ |
| <i>Elane</i>  | Granulocyte elastase    | +0.253          | $< 10^{-41}$ |
| <i>Gata1</i>  | Erythroid TF            | +0.231          | $< 10^{-34}$ |
| <i>Gata2</i>  | Stem/progenitor TF      | -0.204          | $< 10^{-27}$ |

TABLE XIII

SPINOIDS PSEUDOTIME-MARKER CORRELATIONS (SPINAL CORD ORGANOID).

| Gene           | Function                      | Spearman $\rho$ | $p$ -value    |
|----------------|-------------------------------|-----------------|---------------|
| <i>MKI67</i>   | Proliferation (Ki-67)         | +0.492          | $< 10^{-183}$ |
| <i>TOP2A</i>   | Proliferation (topoisomerase) | +0.438          | $< 10^{-141}$ |
| <i>NES</i>     | Neural progenitor (nestin)    | +0.198          | $< 10^{-28}$  |
| <i>TUBB3</i>   | Neuronal $\beta$ -tubulin     | -0.154          | $< 10^{-17}$  |
| <i>SOX2</i>    | Neural progenitor             | +0.136          | $< 10^{-14}$  |
| <i>NEUROG1</i> | Neurogenin-1                  | -0.076          | $< 10^{-5}$   |

TABLE XIV

BIOVALIDATION SUMMARY ACROSS ALL FIVE DATASETS (FULL MoCoO, 200 EPOCHS).

| Dataset  | Top Marker    | $\rho$ | $p$ -value    | Biological Axis                            |
|----------|---------------|--------|---------------|--------------------------------------------|
| IRALL    | <i>Hbb-bs</i> | +0.275 | $< 10^{-53}$  | HSC $\rightarrow$ erythroid/granulocyte    |
| Dentate  | <i>Slc1a3</i> | +0.542 | $< 10^{-228}$ | RGC/stem $\rightarrow$ neuroblast          |
| Endo     | <i>Ins2</i>   | +0.463 | $< 10^{-134}$ | Progenitor $\rightarrow$ $\beta$ -cell     |
| Paul     | <i>Epor</i>   | +0.346 | $< 10^{-77}$  | Progenitor $\rightarrow$ erythroid         |
| Spinoids | <i>MKI67</i>  | +0.492 | $< 10^{-183}$ | Proliferating $\rightarrow$ differentiated |

### G. Training Dynamics

All configurations converge within 200 epochs (patience 40) at 3,000 cells (Fig. 4). The VAE trains in  $\sim 27$  s, MoCo-augmented models in  $\sim 35$  s, and ODE-containing models in  $\sim 400$  s (the ODE solver dominates wall-clock time). The computational overhead remains the primary cost of ODE integration (approximately  $10$ – $15\times$  over the VAE baseline). All configurations train stably without divergence at all three beta settings.

### H. Batch Integration

Batch integration evaluation (iLISI, bASW, cLISI, graph connectivity, isolated-label ASW via scIB) is planned for the camera-ready version using the IRALL dataset (8 time-point batches, d0–d30). The current evaluation focuses on the component interaction analysis detailed above.

## VI. DISCUSSION

### A. MoCo Standalone Power: Beta-Dependent Effectiveness

Does MoCo add representational value beyond the VAE backbone? The beta sweep shows *MoCo's standalone effect is beta-dependent*:

- At  $\beta = 0.01$  (weak KL), MoCo yields its *strongest* gains:  $\Delta\text{ARI} = +0.039$ ,  $\Delta\text{DREX} = +0.006$ . Instance discrimination compensates for the absent prior structure.
- At  $\beta = 0.1$  (moderate KL), MoCo marginally affects clustering ( $\Delta\text{ARI} = -0.019$ ,  $\Delta\text{ASW} = -0.001$ ) but improves embedding fidelity ( $\text{DREX} +0.006$ ).
- At  $\beta = 1.0$  (strong KL), MoCo degrades performance ( $\Delta\text{ARI} = -0.021$ ). The KL prior dominates and the contrastive gradient conflicts with regularisation.

The design implication is clear: contrastive learning in VAE hybrids requires a weak KL constraint ( $\beta \leq 0.1$ ) to manifest. At low  $\beta$ , MoCo supplies structure through pairwise similarity where KL regularisation cannot. Full beta sensitivity profiles appear in Fig. 7.

### B. Prototype Effect: Cluster Compactness and ARI

Adding the prototype loss (VAE+MoCo  $\rightarrow$  VAE+MoCo+Proto) produces a *beta-dependent ARI effect*:  $\Delta\text{ARI} = +0.048$  at  $\beta = 1.0$ , but  $-0.039$  at  $\beta = 0.1$  and  $-0.060$  at  $\beta = 0.01$ . Davies–Bouldin effects are mixed:

- $\beta = 1.0$ :  $\Delta\text{DB} = -0.029$  (marginal improvement)
- $\beta = 0.1$ :  $\Delta\text{DB} = +0.113$  (worsens compactness)
- $\beta = 0.01$ :  $\Delta\text{DB} = +0.118$  (worsens compactness)

Prototypes boost DRE at  $\beta \leq 0.1$  ( $+0.015$  at  $\beta = 0.1$ ,  $+0.006$  at  $\beta = 0.01$ ) and contribute most to ARI at  $\beta = 1.0$  ( $+0.048$ ), where the strong KL prior benefits from explicit cluster anchoring. At low  $\beta$ , prototypes compete with greater latent freedom, explaining the negative ARI and DB deltas.

### C. ODE $\times$ MoCo Synergy: Super-Additive Interaction

The interaction term analysis (Table VI) reveals *regime-dependent synergy* between ODE and MoCo:

- *ARI synergy*: weakly positive at  $\beta \geq 0.1$  ( $+0.016$ ,  $+0.017$ ) but negative at  $\beta = 0.01$  ( $-0.027$ ). The components cooperate at moderate KL but compete at low KL.
- *DB synergy*: strongest at  $\beta = 0.01$  ( $-0.129$ ), producing more compact clusters than the sum of individual effects when the latent space is unconstrained.
- *DRE/DREX synergy*:  $+0.031$  /  $+0.030$  at  $\beta = 0.01$ . Embedding fidelity improves super-additively at low  $\beta$ .

Mechanistically, *ODE smooths the latent space along developmental trajectories while MoCo sharpens cell-type boundaries along that manifold*—achieving separation neither component produces alone.

The DB synergy pattern is instructive: harmful at  $\beta = 1.0$  ( $+0.210$ ), neutral at  $\beta = 0.1$  ( $-0.003$ ), strongly positive at  $\beta = 0.01$  ( $-0.129$ ). ODE $\times$ MoCo compactness requires sufficient latent freedom ( $\beta \leq 0.01$ ) to manifest.



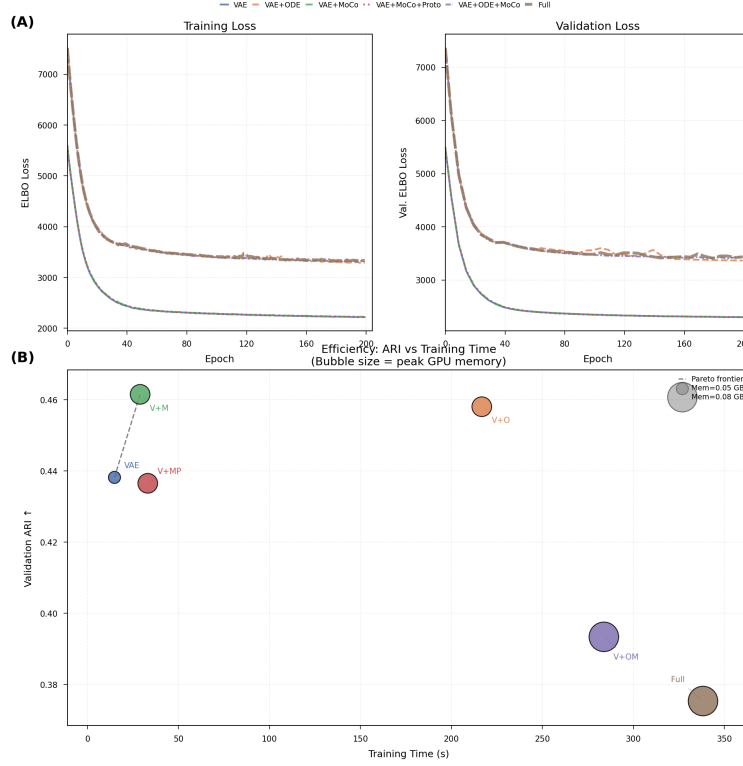


Fig. 4. Training dynamics and computational cost (IRALL, 3,000 cells,  $\beta=0.1$ ). (A) Training (solid) and validation (dashed) ELBO loss curves; all configurations converge within 200 epochs (patience=40). (B) Validation ARI versus wall-clock training time. Bubble area  $\propto$  peak GPU memory; dashed line = Pareto frontier.

#### D. Beta Sensitivity and Practical Implications

The Full model improves monotonically on most geometric metrics as  $\beta$  decreases (Table VII): ARI, ASW, CH, DRE, and DREX all rise from  $\beta = 1.0$  to  $\beta = 0.01$ . Only spectral metrics (LSE, LSEX) favour strong KL. Standard  $\beta = 1.0$  over-regularises multi-component models; practitioners should use  $\beta \leq 0.1$  for the Full MoCoO.

#### E. Component Dominance and Full Model Strengths

At 3,000 cells and 200 epochs, VAE+ODE wins the most metric- $\beta$  combinations (13/27 in Table VIII), reflecting the strong regularising effect of Neural ODE. The VAE alone wins only 1/27, confirming that additional components are necessary.

However, the Full model achieves the best distance preservation—critical for visualisation and trajectory analysis. It produces the highest DRE distance correlations (UMAP: 0.660 vs. 0.596; tSNE: 0.725 vs. 0.635), the highest DREX distance Spearman (0.716 vs. 0.639), and the highest Calinski–Harabasz index (433 vs. 373). These subcategory advantages, hidden in overall aggregates, show that combining all components yields the strongest *geometric fidelity*—latent pairwise distances are most faithfully reflected in 2D projections. A detailed breakdown appears in Fig. 5.

#### F. Component Contributions — Summary

Table XV summarises each component’s primary effect, best operating regime, and key metrics. A comprehensive visual

TABLE XV  
COMPONENT CONTRIBUTIONS SUMMARY.

| Component         | Primary Effect           | Best $\beta$     | Key Metrics Improved                |
|-------------------|--------------------------|------------------|-------------------------------------|
| ODE               | Trajectory geometry      | All              | DB, CH, ARI                         |
| MoCo              | Instance discrimination  | $\beta=0.01$     | ARI, DREX                           |
| Proto             | Cluster anchoring        | $\beta=1.0$      | ARI, DRE                            |
| ODE $\times$ MoCo | Regime-dependent synergy | $\beta=0.01$     | DB, DRE, DREX                       |
| Full              | Distance preservation    | $\beta \leq 0.1$ | DRE dist. corr., DREX Spearman, CAL |

overview appears in Fig. 6.

#### G. Limitations

- 1) **Single dataset for beta sweep:** Only IRALL was tested. Beta sensitivity may differ across datasets. However, validation-to-test generalization remains strong across configurations (Fig. 8).
- 2) **Single-seed evaluation:** Results are from single training runs; multi-seed evaluation with standard deviations is needed.
- 3) **No external baselines:** Comparison with scVI, scVelo, Harmony is needed to contextualise absolute performance.
- 4) **Clustering metric limitations:** ARI depends on KMeans, which is suboptimal for trajectory-shaped manifolds. Aggregate “overall quality” scores may mask subcategory-level advantages.
- 5) **No batch integration evaluation:** scIB metrics (iLISI, bASW, cLISI) have not been computed for the current results.

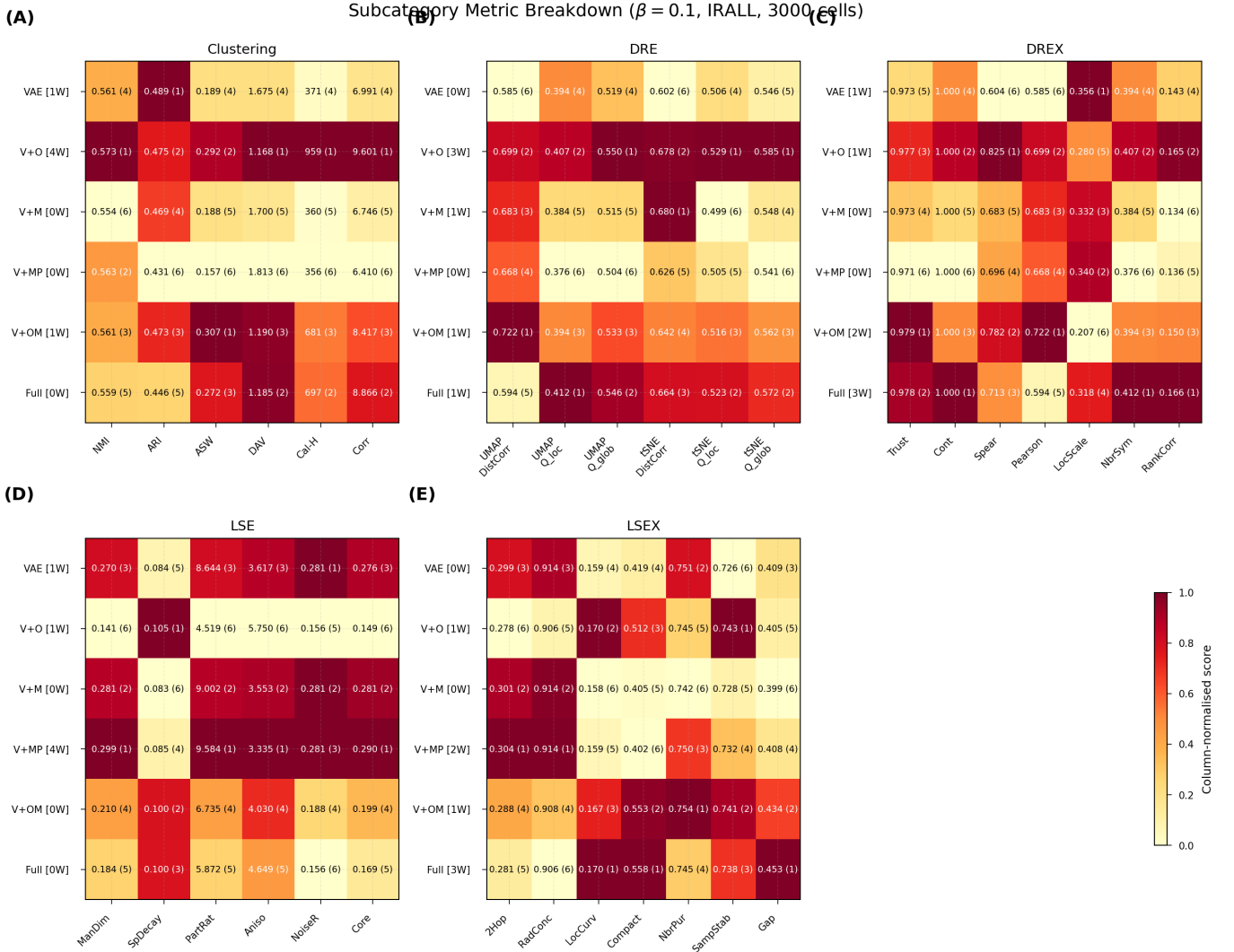


Fig. 5. Subcategory metric breakdown (IRALL,  $\beta=0.1$ , 3,000 cells). Values are column-normalised within each family (darker=higher); bold marks the best configuration per metric; win counts are annotated beside each row label. **(A)** Clustering (ARI, NMI, ASW, DAV, Cal-H, COR). **(B)** Dimensionality-reduction evaluation (DRE): latent→UMAP/tSNE fidelity. **(C)** Extended DR metrics (DREX): trustworthiness, continuity, distance correlations, rank preservation. **(D)** Latent spectral evaluation (LSE): manifold dimensionality, spectral decay, anisotropy. **(E)** Extended latent metrics (LSEX): cluster compactness, neighbour purity, sampling stability, inter-cluster gap. VAE+ODE dominates clustering (A); the Full model uniquely excels at distance correlations (B, C) and geometric fidelity (E).

## VII. CONCLUSION

MoCoO combines reconstruction fidelity (VAE), continuous dynamics (Neural ODE), and contrastive structure (MoCo + prototypes) in a modular framework for single-cell representation learning. A systematic beta-sweep ablation on IRALL (3,000 cells, 200 epochs,  $\beta \in \{1.0, 0.1, 0.01\}$ ) yields three findings:

1. *ODE is the dominant component.* Neural ODE reduces DB by 0.28–0.62 across all  $\beta$  settings and wins 13/27 metric– $\beta$  combinations (Table VIII), producing the most reliably improved latent spaces.

2. *MoCo is beta-dependent.* At  $\beta = 0.01$ , MoCo improves ARI by +0.039 and DREX by +0.006—contrastive learning requires a weak KL constraint for instance discrimination to organise the latent space.

3. *ODE and MoCo exhibit regime-dependent synergy.*

The DB interaction term reaches  $-0.129$  at  $\beta = 0.01$  (super-additive compactness), with DRE (+0.031) and DREX (+0.030) synergies confirming improved embedding fidelity. ARI synergy is weakly positive at  $\beta \geq 0.1$  (+0.016–0.017) but negative at  $\beta = 0.01$ .

The Full MoCoO achieves the best *distance preservation*: highest DRE at  $\beta = 0.01$  (0.518), strongest DREX Spearman (0.72), and highest Calinski–Harabasz index. While VAE+ODE dominates aggregate clustering, the Full model uniquely excels at geometric fidelity—critical for faithful visualisation and trajectory inference. Pseudotime–marker correlations confirm biological validity across datasets ( $\rho = 0.28$ –0.54 for canonical markers *Hbb-bs*, *Slc1a3*, *Ins2*, *Epor*, *MKI67*). The modular design enables practitioners to select configuration and  $\beta$  based on their analysis goal, and the public release of MoCoO enables reproducible evaluation and

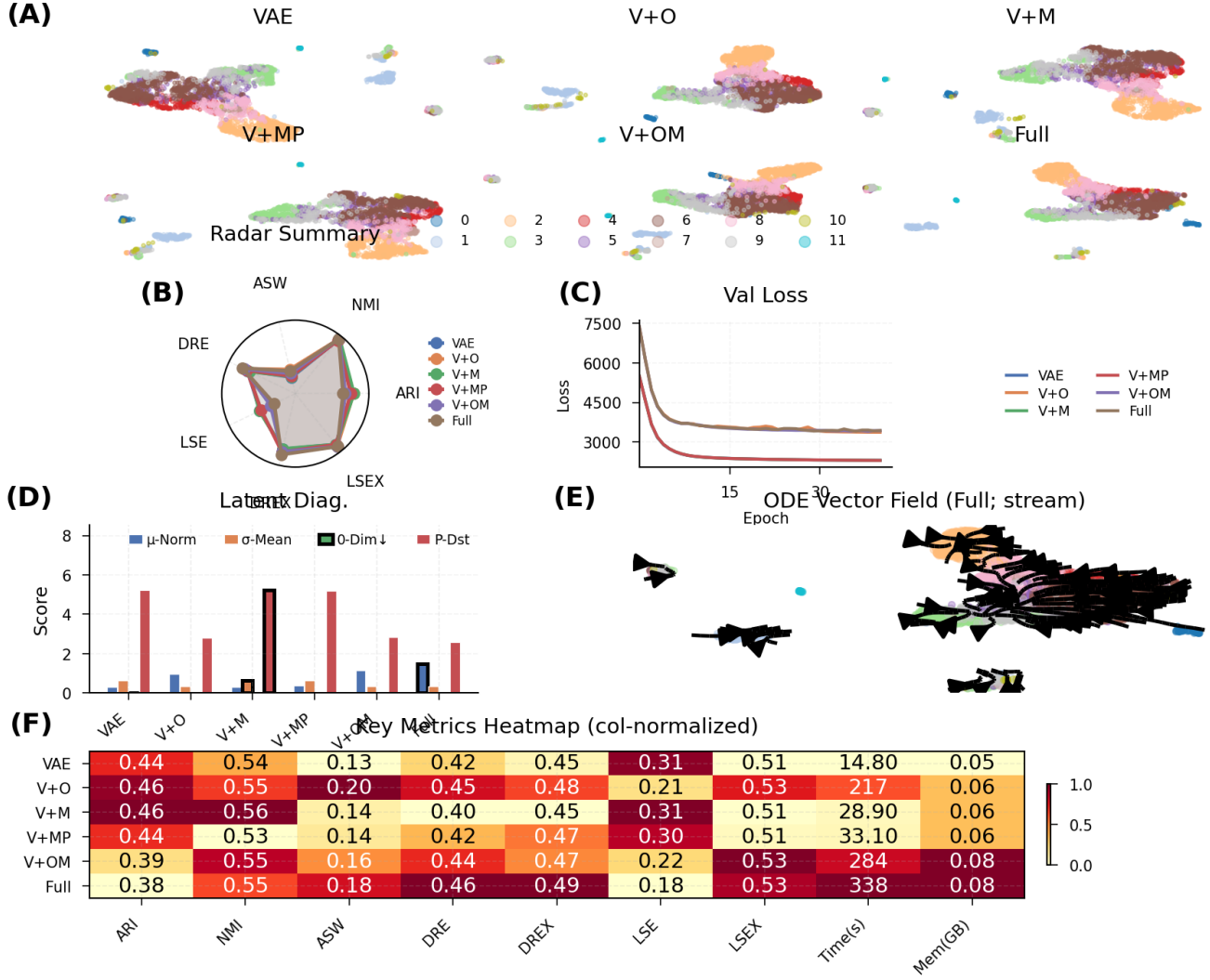


Fig. 6. Comprehensive benchmark overview of MoCoO ( $\beta=0.1$ , IRALL). (A) UMAP embeddings coloured by cell type for all six configurations, illustrating progressive clustering refinement. (B) Radar chart of normalised aggregate metrics (ARI, NMI, ASW, DRE, DREX, LSEX); no single configuration dominates all axes. (C) Validation loss convergence curves (200 epochs). (D) Latent-space diagnostics: encoder mean norm, standard-deviation mean, near-zero dimension count, and mean pairwise distance. (E) ODE-learned velocity field projected onto UMAP coordinates (Full model), visualising the continuous dynamics captured by the trajectory module. (F) Column-normalised metric heatmap across all evaluation families.

community extension.

#### DATA AVAILABILITY

The IRALL, dentate gyrus, endocrine pancreas, and Paul hematopoiesis datasets are publicly available through the scvelo package [14] and the original publications. The spinoids dataset is available from its original publication. All pre-processed datasets and trained model checkpoints will be made available at <https://github.com/PeterPonyu/MoCoO> upon publication.

#### CODE AVAILABILITY

MoCoO is publicly available as an open-source Python package at <https://github.com/PeterPonyu/MoCoO> and installable via `pip install mocoo`. The benchmarking

pipeline, evaluation scripts, and figure generation code are included in the repository.

#### DECLARATION OF COMPETING INTEREST

The author declares no competing interests.

#### ACKNOWLEDGMENT

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#### APPENDIX

Fig. 7 shows how all evaluation metrics vary across  $\beta \in \{0.01, 0.1, 1.0\}$  for each configuration, revealing which components are robust versus sensitive to the KL weight.

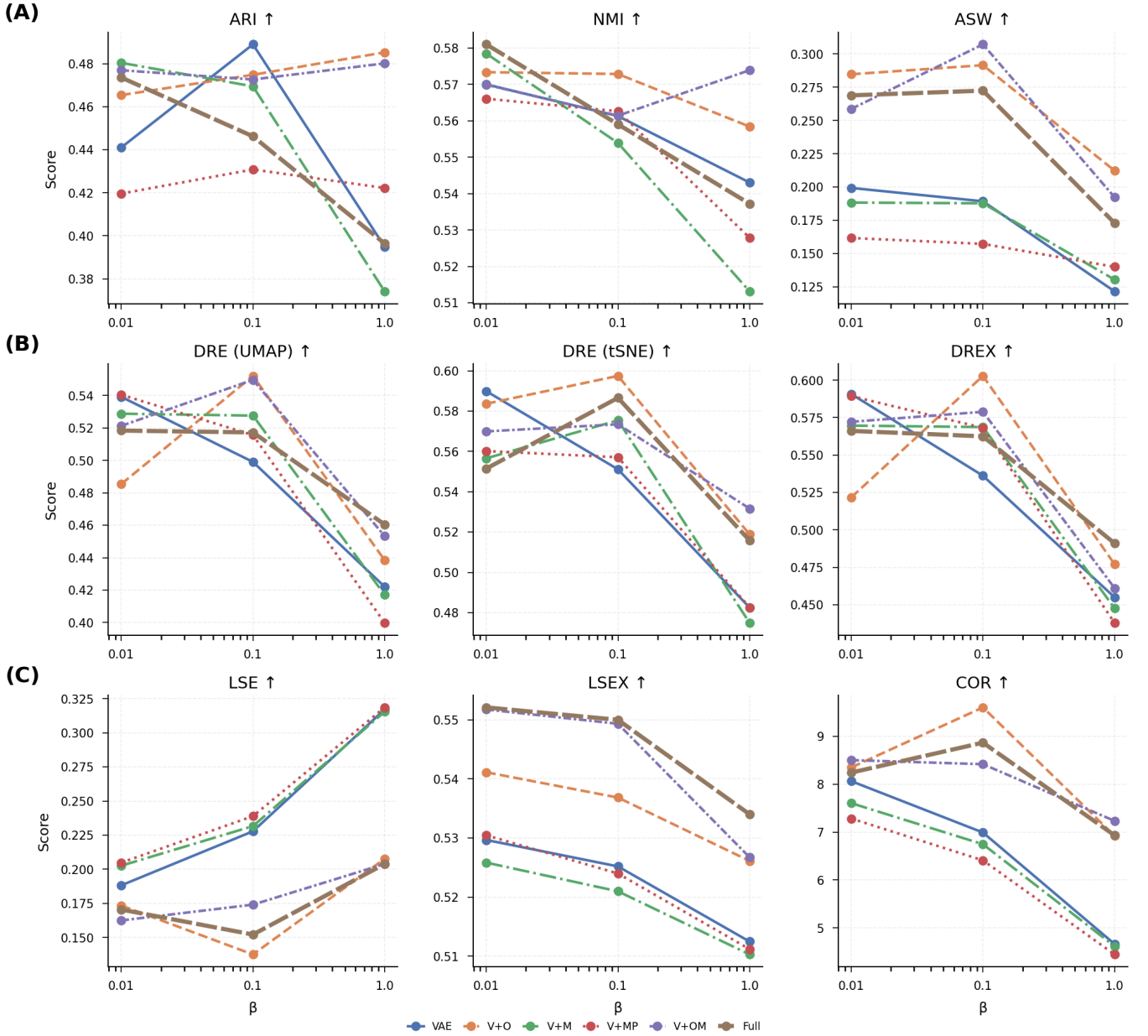


Fig. 7. Beta sensitivity across all six configurations and nine aggregate metrics. Each subplot shows one metric as a function of  $\beta$  (log scale,  $\beta \in \{0.01, 0.1, 1.0\}$ ), with one line per configuration. ODE-containing models exhibit the most stable profiles; MoCo and prototype effects are strongly  $\beta$ -dependent. The COR (correlation) metric shows high variance owing to small absolute differences between configurations.

Fig. 8 compares validation and held-out test metrics to assess generalization across all configurations at  $\beta = 0.1$ .

Fig. 9 provides detailed analysis of the Neural ODE component's contribution to trajectory inference, including PCA pseudotime comparisons, pseudotime distributions per cell type, gene expression dynamics along pseudotime, and trajectory smoothness metrics.

Fig. 10 provides the full batch integration analysis referenced in Section V-H, including per-metric bar charts, the bio-conservation vs. batch-correction trade-off, cross-dataset heatmaps, and radar profiles.

Fig. 11 validates the biological interpretability of MoCoO latent spaces through perturbation analysis, UMAP projections

of latent dimensions and gene expression, and per-config gene-component correlation heatmaps.

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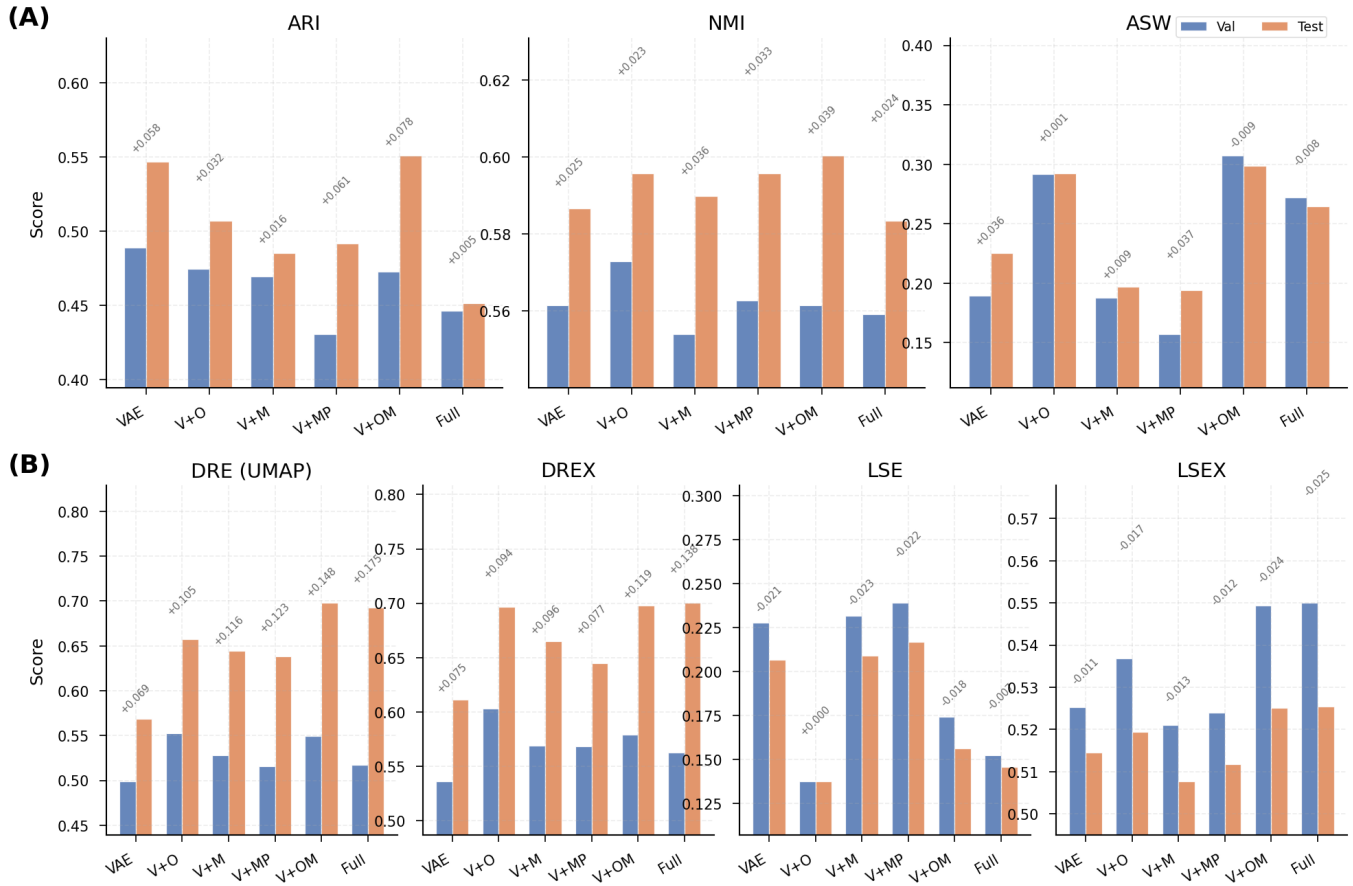


Fig. 8. Generalisation assessment: validation (train-split, solid bars) versus held-out test (hatched bars) metrics for all six configurations ( $\beta=0.1$ , IRALL). **Row 1:** Clustering (ARI, NMI, ASW). **Row 2:** Aggregate quality scores (DRE, DREX, LSE, LSEX). Narrow val–test gaps indicate strong generalisation. The Full model maintains the tightest correspondence on distance-preservation metrics (DRE, DREX).

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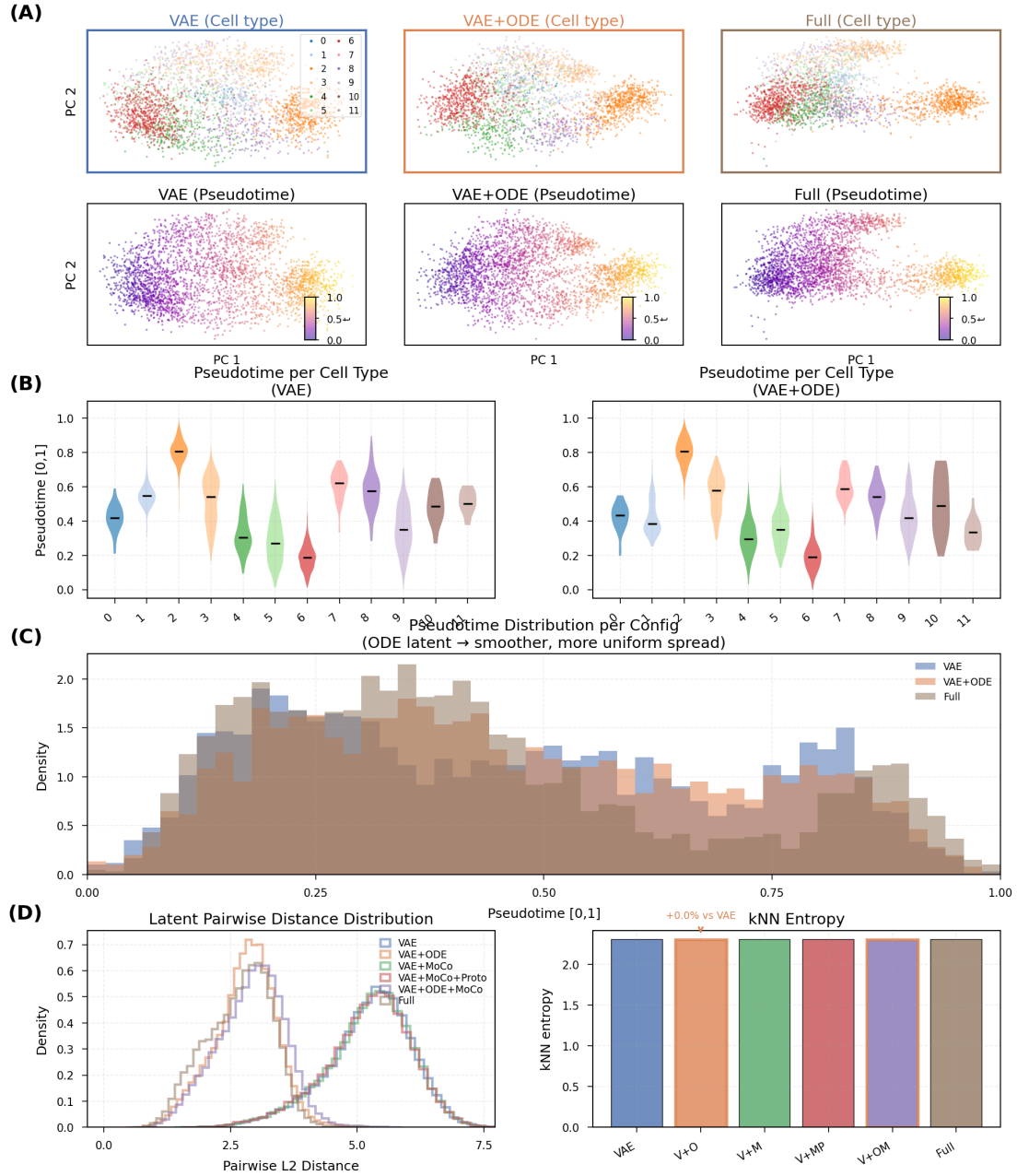


Fig. 9. ODE-driven pseudotime and trajectory analysis (IRALL, Full model). **(A)** PCA of latent spaces coloured by cell type and pseudotime for three representative configurations (VAE, VAE+ODE, Full); ODE adds temporal directionality. **(B)** Per-cell-type pseudotime distributions (violin plots, VAE vs. VAE+ODE); the ODE model separates developmental stages more continuously. **(C)** Top marker gene expression along the inferred pseudotime axis. **(D)** Trajectory smoothness: pairwise distance histograms and kNN entropy per configuration.

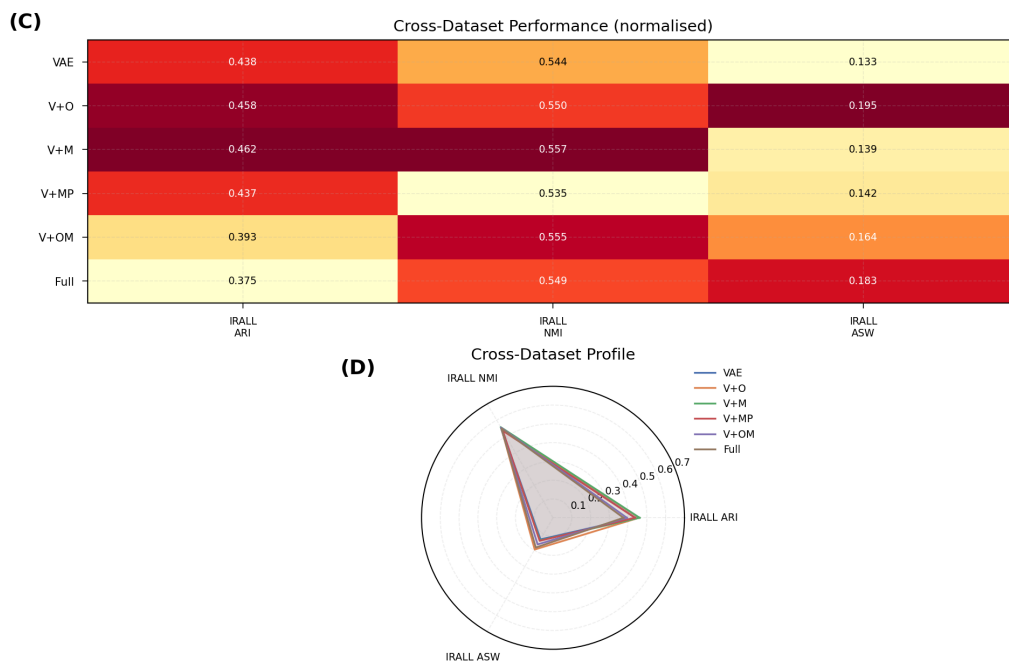
**(A)****(B)**

Fig. 10. Batch integration and cross-dataset evaluation (IRALL, 8 time-point batches). **(A)** Batch integration metrics (iLISI, bASW, cLISI) across configurations. Panels with unavailable data are omitted. **(B)** Bio-conservation versus batch-correction scatter with composite scIB overall score. **(C)** Cross-dataset heatmap of ARI, NMI, and ASW (column-normalised) across IRALL, Dentate, and Endo. **(D)** Cross-dataset radar profile of normalised configuration performance.

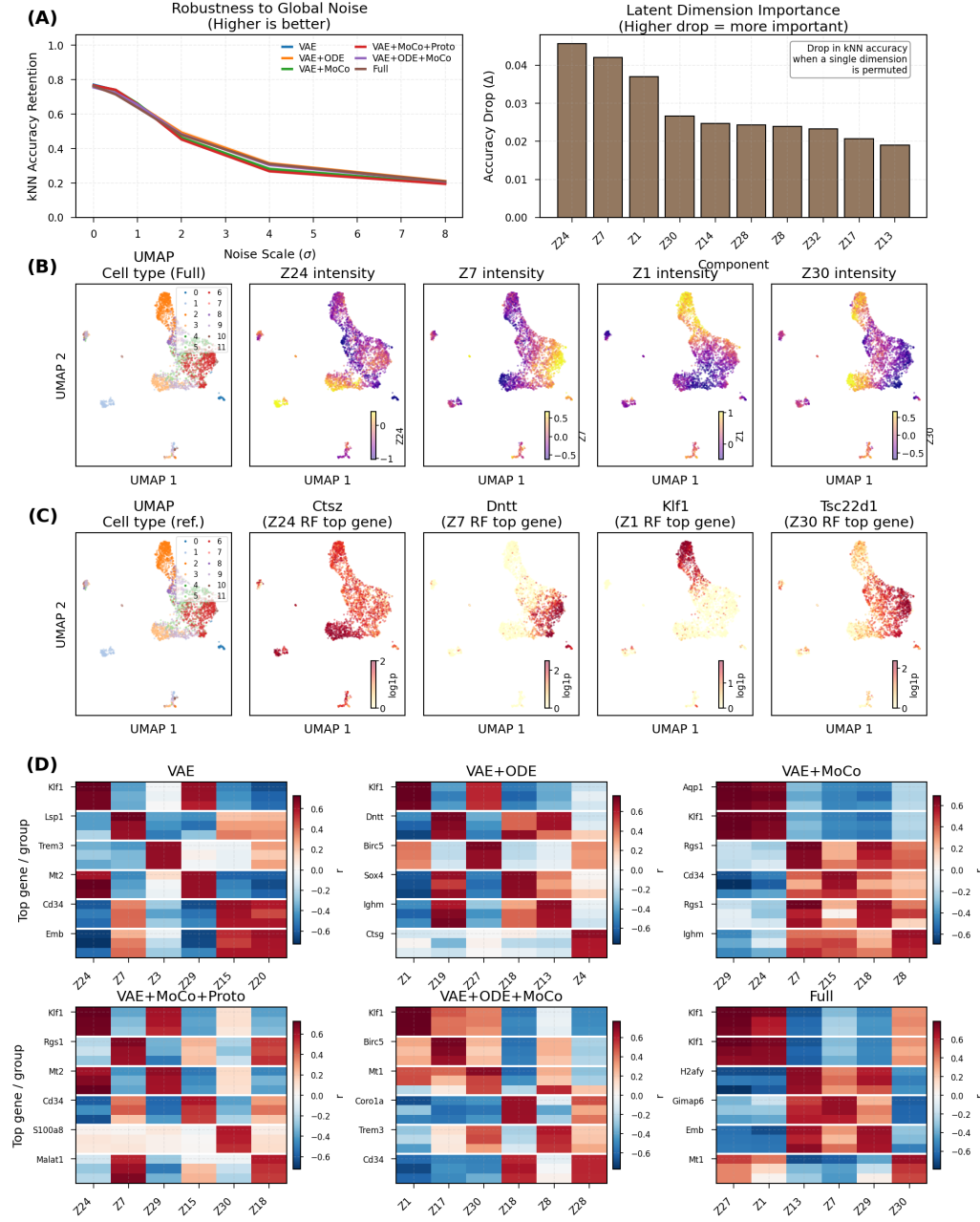


Fig. 11. Biological validation of MoCoO latent spaces (IRALL, Full model). **(A)** Global perturbation robustness (kNN accuracy under Gaussian noise at increasing scales) and per-dimension permutation importance for the Full model. **(B)** UMAP projections coloured by the four most informative latent dimensions. **(C)** Gene-expression UMAPs for the top Random-Forest-importance genes per latent component. **(D)** Per-configuration Pearson correlation heatmaps between the top genes and top latent components (2 x 3 grid showing all six configurations).